AGT1 - MODULE-3

Interactive visualization at: https://mehrsachal.github.io/InteractiveLearning/#/surface-interpolation-companion

Objectives

- Understand the difference between discrete and continuous data.
- Understand the mechanics of Surface Interpolation.
- Use the Spline Tool (Regularized).
- Use the Spline Tool (Tension).

Deliverables (To be submitted)

- Map showing use of each IDW tool. (Screenshot **not** Map Layout)
- Map showing use of both spline tools. (Screenshot **not** Map Layout)
- Map showing use of Kriging tool. (Screenshot **not** Map Layout)
- Map showing use of IDW tool with barrier. (Screenshot **not** Map Layout)
- Map showing use of spline with barrier tool. (Screenshot **not** Map Layout)

Lab Settings

Use Data in folder: **AGT1-Data/Module3**

Scenario

In plain terms, "interpolation" is simply making a highly educated guess about missing information by looking at the information you already have.

Here is the easiest way to think about it: Imagine you are climbing a mountain.

- At the **bottom**, your thermometer says it is 20°C.
- At the **top**, your thermometer says it is 10°C.
- You didn't check the temperature exactly **halfway** up, but if someone asked you to guess what it was, what would you say?

You would probably guess 15°C.

You just *interpolated*. You used two known data points (top and bottom) to mathematically guess the missing data point in the middle.

The temperature example above is a **straight line** between two points. In GIS, you aren't just guessing in a straight line; you are guessing across an entire landscape in every direction. It's called **surface interpolation.**

What is Surface Interpolation?

Imagine you stick a bunch of tent poles of different heights into the ground. Now, imagine throwing a giant, stretchy rubber sheet over the top of them. The sheet stretches and bends to touch the top of every single pole, filling in all the empty air between them.

In GIS, the **"tent poles"** are the scattered elevation **points** you actually measured in the field. The **"rubber sheet"** is the continuous terrain surface the computer generates to guess what the ground looks like everywhere else.

This mathematical magic of "filling in the blanks" between your known points is called **Surface Interpolation**.

In this exercise we shall explore different interpolation methods.

IDW

Imagine you want to guess the temperature in your backyard, but you only have thermometers at your neighbor's houses.

Inverse Distance Weighting (IDW) is basically a "weighted guessing" game. It works on one simple rule: **the closer a neighbor is, the more you trust their number.**

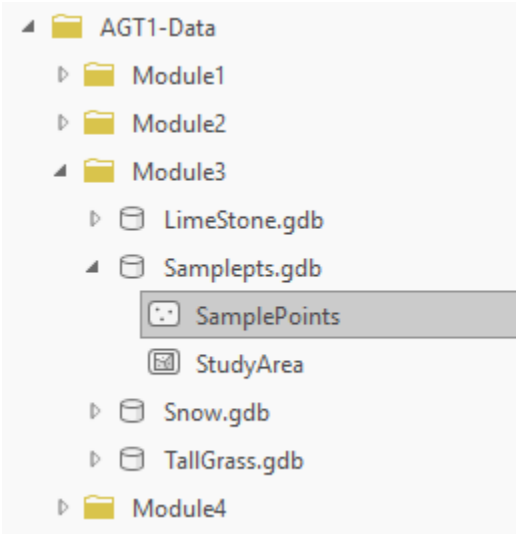
- If your next-door neighbor says it's 30°C, that carries a lot of weight.
- If a neighbor three streets away says it's 20°C, you still listen to them, but you don't trust them as much because they are further away.

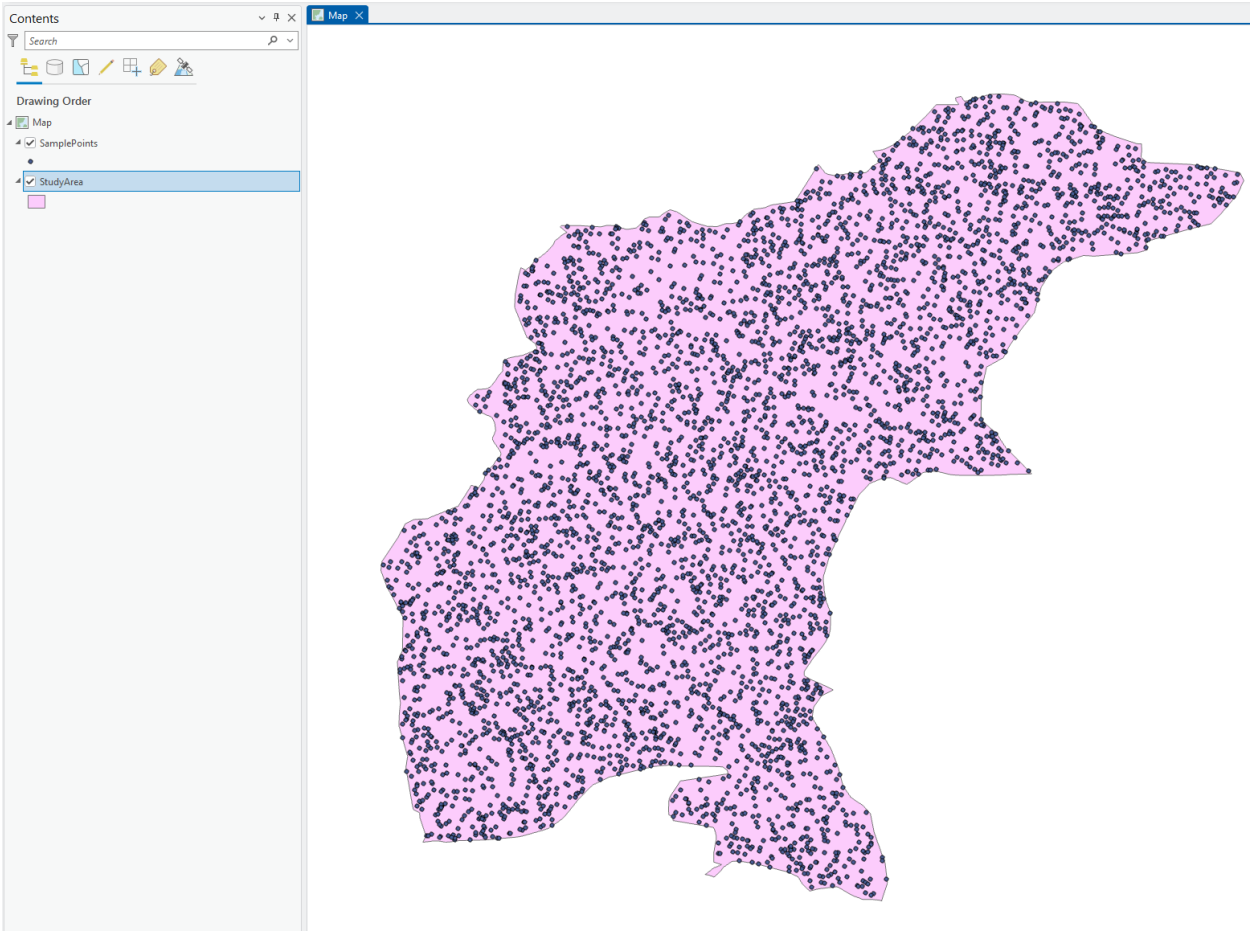
To get your final answer, you take all those nearby numbers, give the "close" ones a big vote and the "far" ones a tiny vote, and find the average. It's a great way to fill in the blanks on a map, making sure the areas between your known points look smooth and logical.

Goal: Use IDW tool.

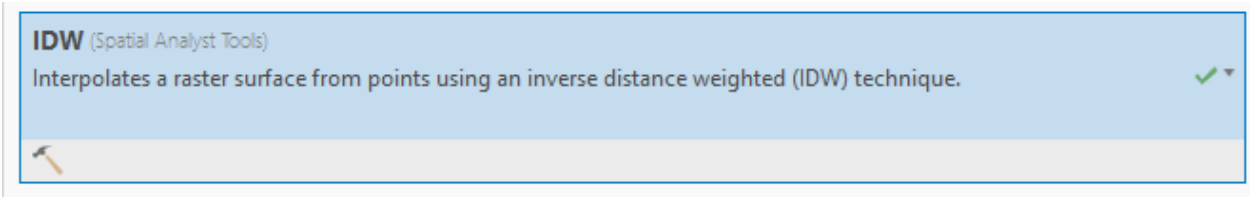
Scenario: We know the elevation of a sample of points in our study area. We need to use it to generate an elevation raster for the entire area.

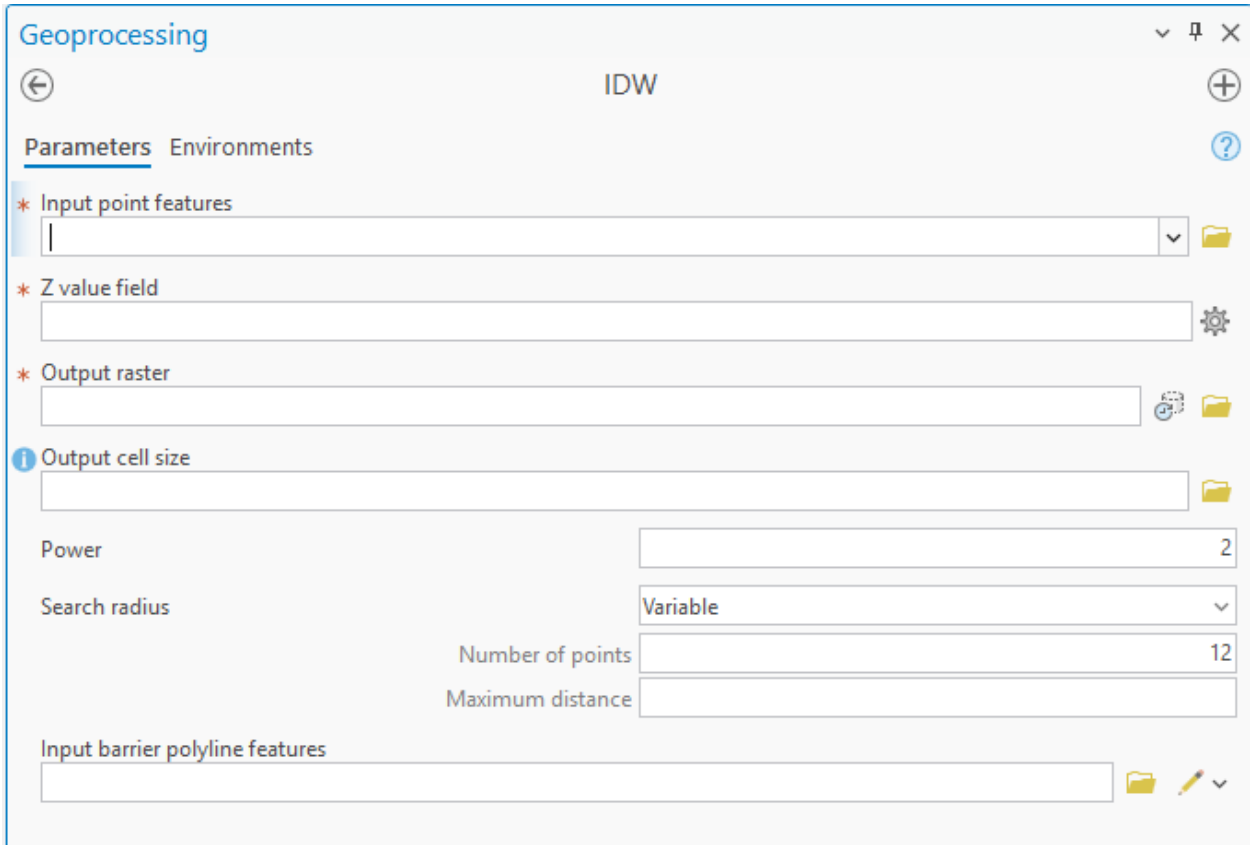
Load **SamplePoints** and **StudyArea:**





Open IDW Tool (Spatial Analyst):





Required Inputs

- **Input point features:** These are your "known" data points—the locations where you already have measurements (like weather stations or soil samples).
- **Z value field:** This is the specific column in your data that contains the number you want to map (e.g., elevation, temperature, or pollution levels).
- **Output raster:** The name and location where your new, smooth map will be saved.

Key Controls

- **Output cell size:** This determines the resolution. A smaller number creates a finer, high-quality map with more detail, while a larger number creates a pixelated, blocky map.
- **Power:** This controls the "drop-off" of influence.

    - **High Power (e.g., 3 or 4):** Only very close points matter. Your map will look more "peaked" or spotted.
    - **Low Power (e.g., 1):** Faraway points get a bigger say. Your map will look smoother and more averaged out.

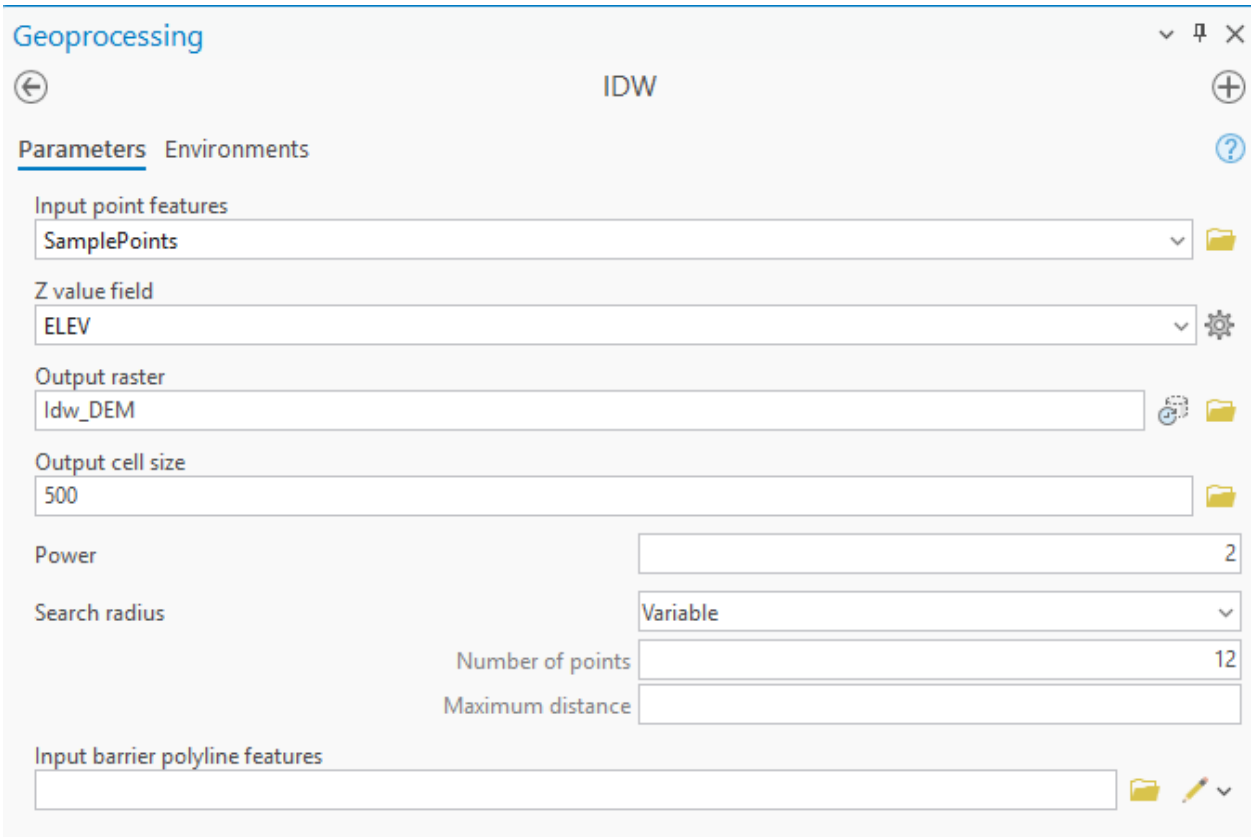
- **Search radius:** This tells the tool how far to look for neighbors.

- **Variable:** It will keep looking until it finds a specific number of neighbors (default is 12).
- **Fixed:** It only looks within a specific distance (e.g., 5km), no matter how many points are there.

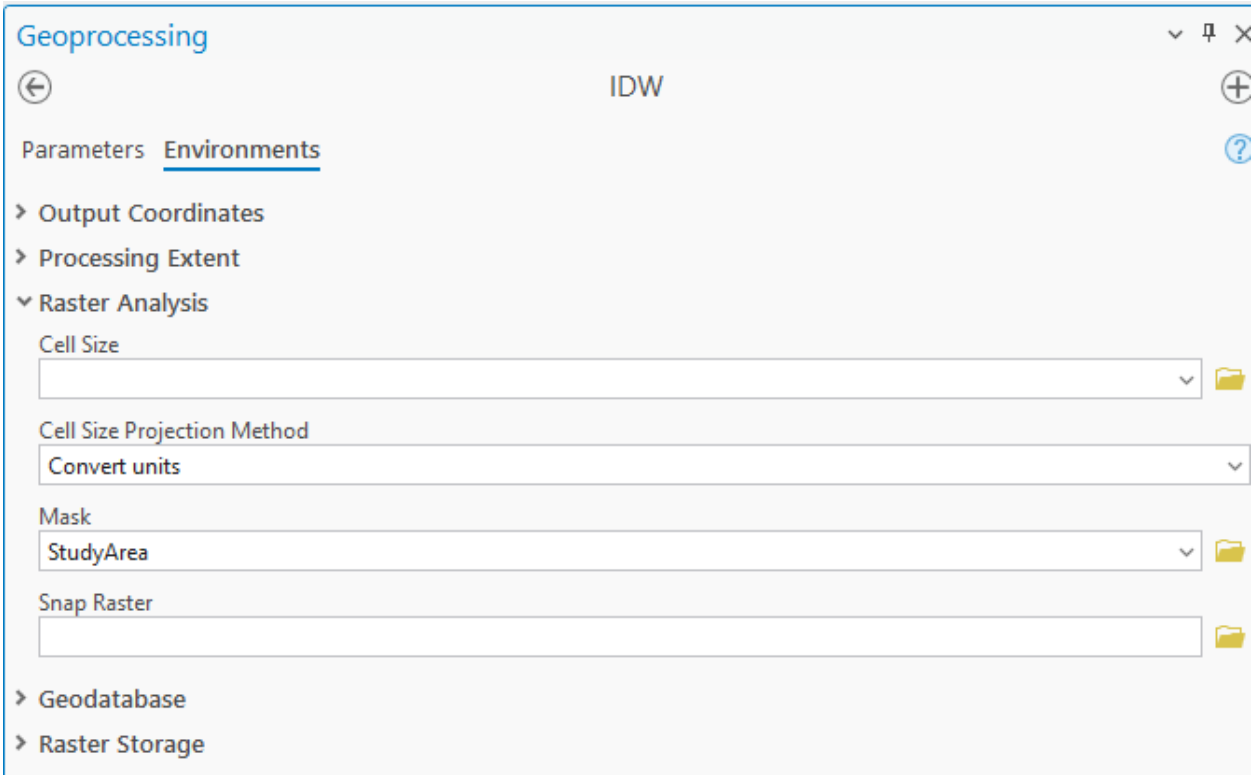
Advanced

- **Input barrier polyline features:** Think of this as a "wall" (like a cliff or a river). If a point is on the other side of this line, the tool will ignore it, preventing data from "bleeding" across physical boundaries where it shouldn't.

Configure it as follows:



Before Processing we must set a **Mask** in the tool's environment:



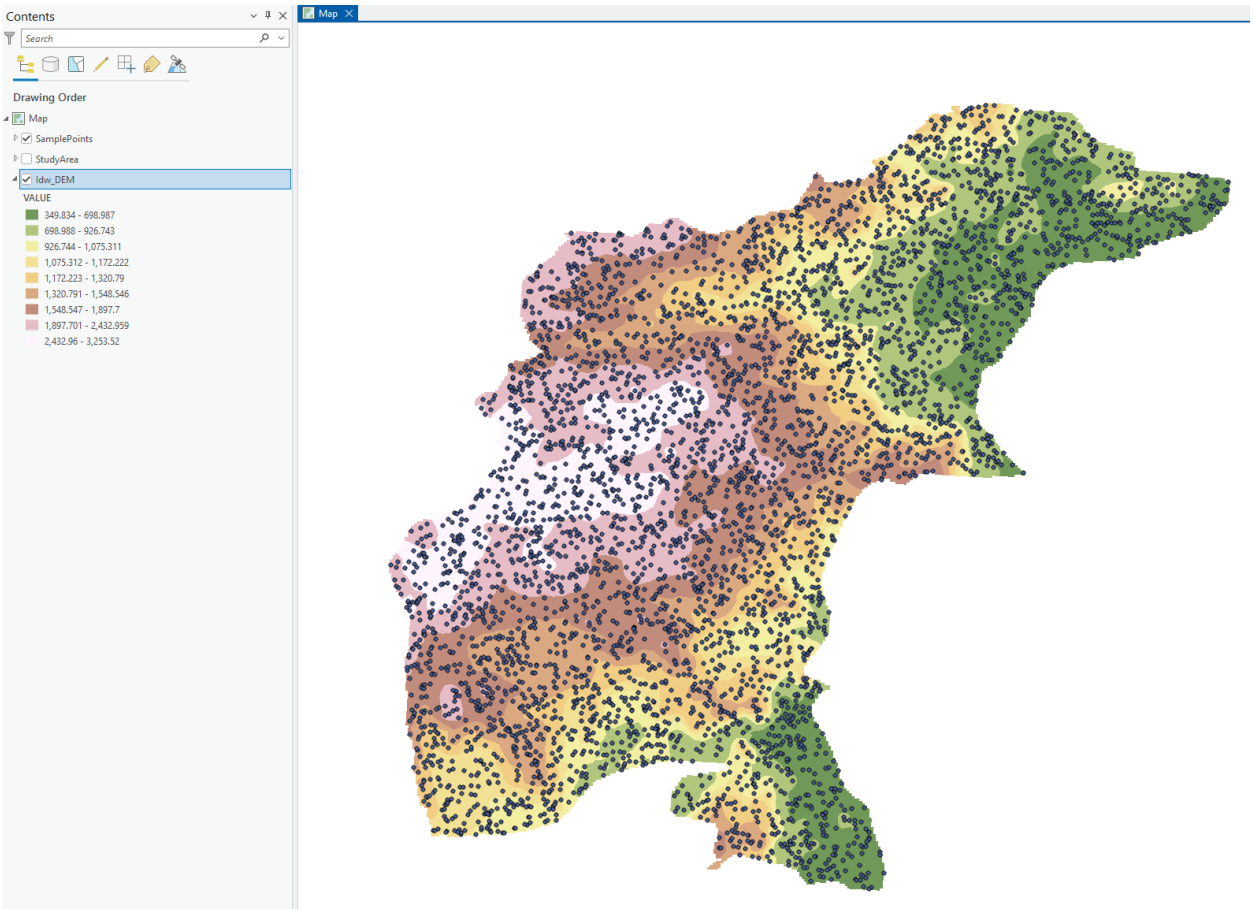
A **Mask** is essentially a "cookie cutter" for your map.

While the parameters tell the tool *how* to calculate values, the Mask setting in tells it *where* to stop. It prevents the tool from creating a giant rectangle of data, instead "clipping" the output to the exact shape of your study area.

This is necessary to:

- **Keep things tidy:** It removes messy data "bleeding" outside your specific borders.
- **Save space:** It ensures areas outside your boundary are marked as "NoData" rather than being filled with useless estimates.
- **Match layers:** It makes sure your results align perfectly with other map layers, like city or forest boundaries.

Hit **Run:**


Contents
Search
Drawing Order
Map
SamplePoints
StudyArea
Idw_DEM
VALUE
349.834 - 698.987
698.988 - 926.743
926.744 - 1,075.311
1,075.312 - 1,172.222
1,172.223 - 1,320.79
1,320.791 - 1,548.546
1,548.547 - 1,897.7
1,897.701 - 2,432.959
2,432.96 - 3,253.52
Map

Spline Tool

The **Spline** tool is a deterministic interpolation method that fits a flexible, mathematical surface through your data points, much like a thin sheet of rubber being pulled to touch every point.

Unlike **IDW**, which can look "bumpy," Spline is designed to create the smoothest possible surface with continuous curvature. ArcGIS offers two primary versions:

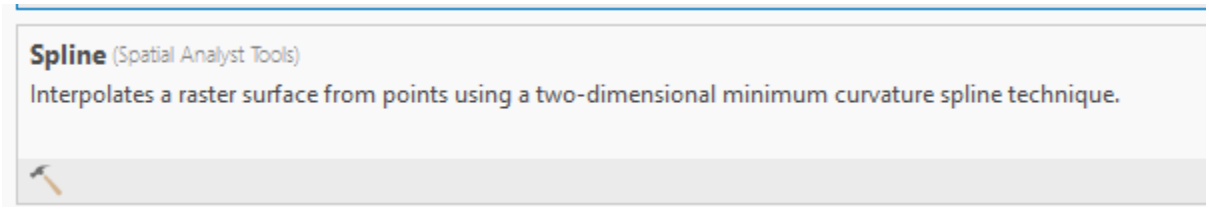
- **Regularized:** This version results in a smooth, gradually changing surface. It allows the values of the output surface to potentially exceed the range of the input data, which is useful for creating gentle terrain or identifying subtle trends.
- **Tension:** This version constrains the surface more tightly to the data points, acting like a stiffer material. The resulting surface is flatter and the values are generally kept closer to the input range, making it ideal for data that shouldn't have extreme peaks or valleys.

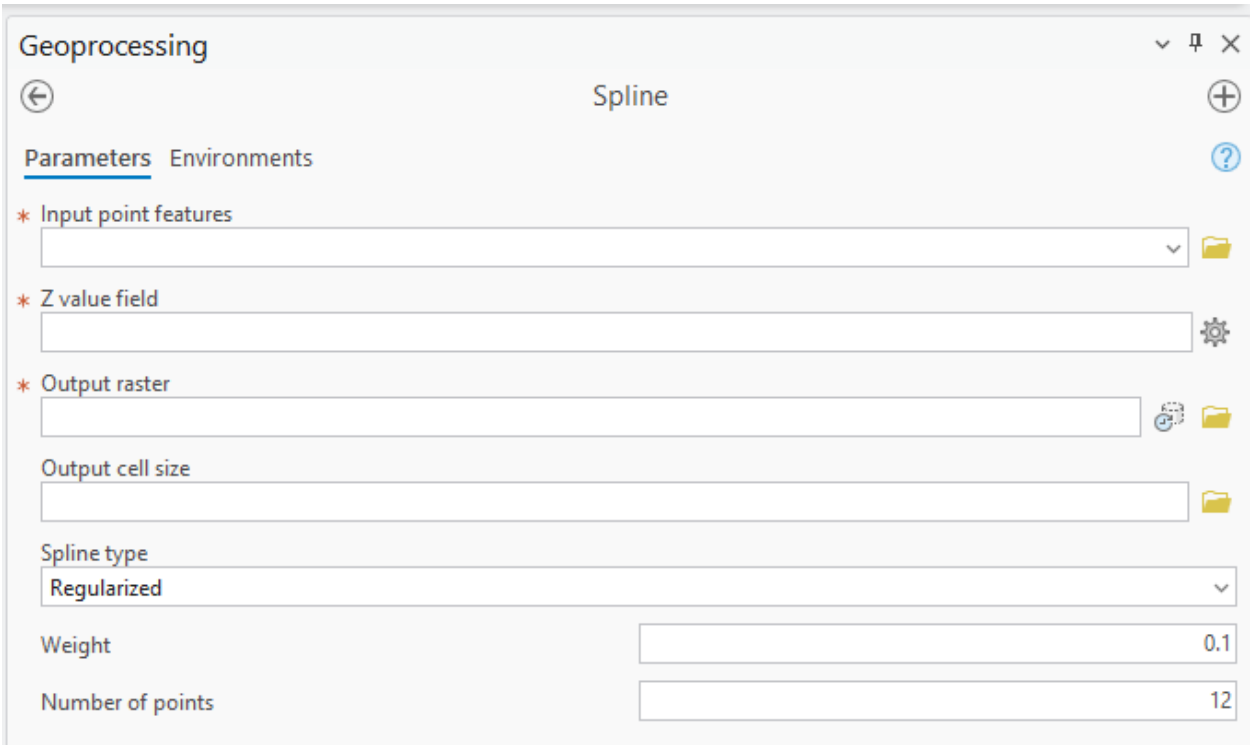
Goal: Use Spline tool.

Scenario: We know the elevation of a sample of points in our study area. We need to use it to generate an elevation raster for the entire area.

Load **SamplePoints** and **StudyArea:**

Open **Spline Tool:**



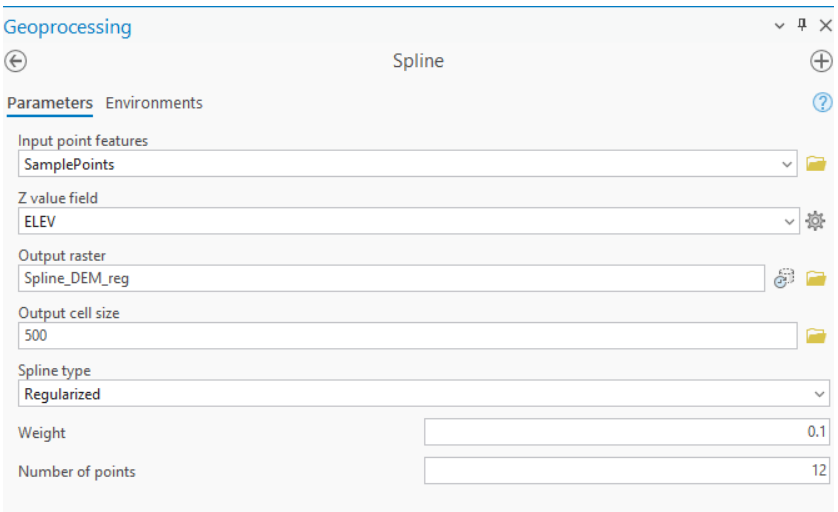


- **Input point features:** These are the measured locations you are using to build the surface.
- **Z value field:** The specific attribute being interpolated (e.g., elevation or rainfall).
- **Output raster:** Where the final smooth surface will be saved.
- **Output cell size:** Determines the resolution of the final map (smaller numbers mean higher detail).

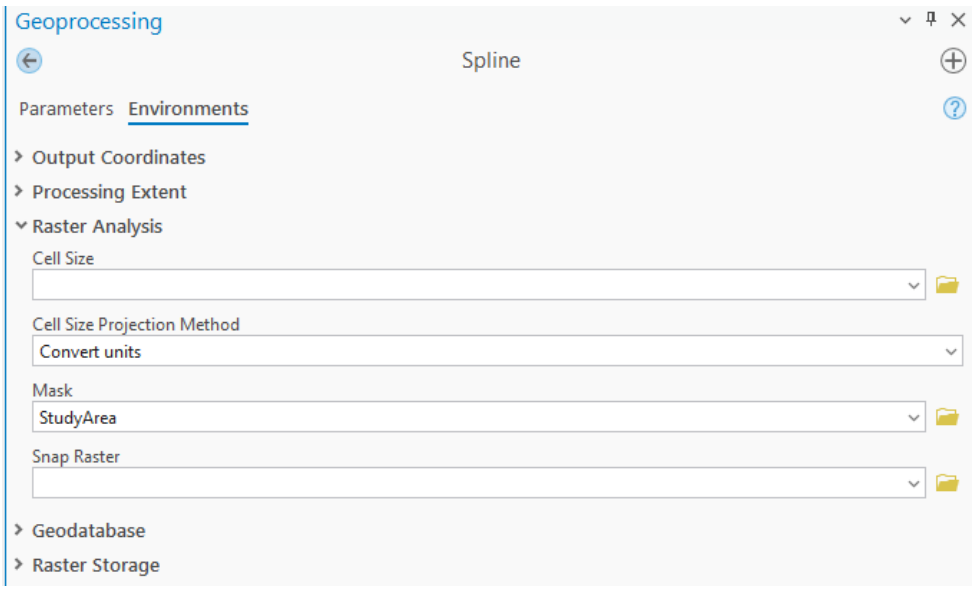
- **Spline type:**
    - **Regularized:** Creates a very smooth, rounded surface. It is best for things like elevation where you want a natural, flowing look.
    - **Tension:** Makes the surface "stiffer." It keeps the predicted values closer to the actual data points and prevents the surface from overshooting or creating wild curves.

- **Weight:** This defines the influence of the chosen spline type.
    - For **Regularized**, a higher weight creates a smoother surface.
    - For **Tension**, a higher weight makes the surface more rigid and flat.
- **Number of points:** This specifies how many nearby points the tool uses to calculate the shape of the curve for each specific area. Using more points generally leads to a smoother overall result but requires more processing power.

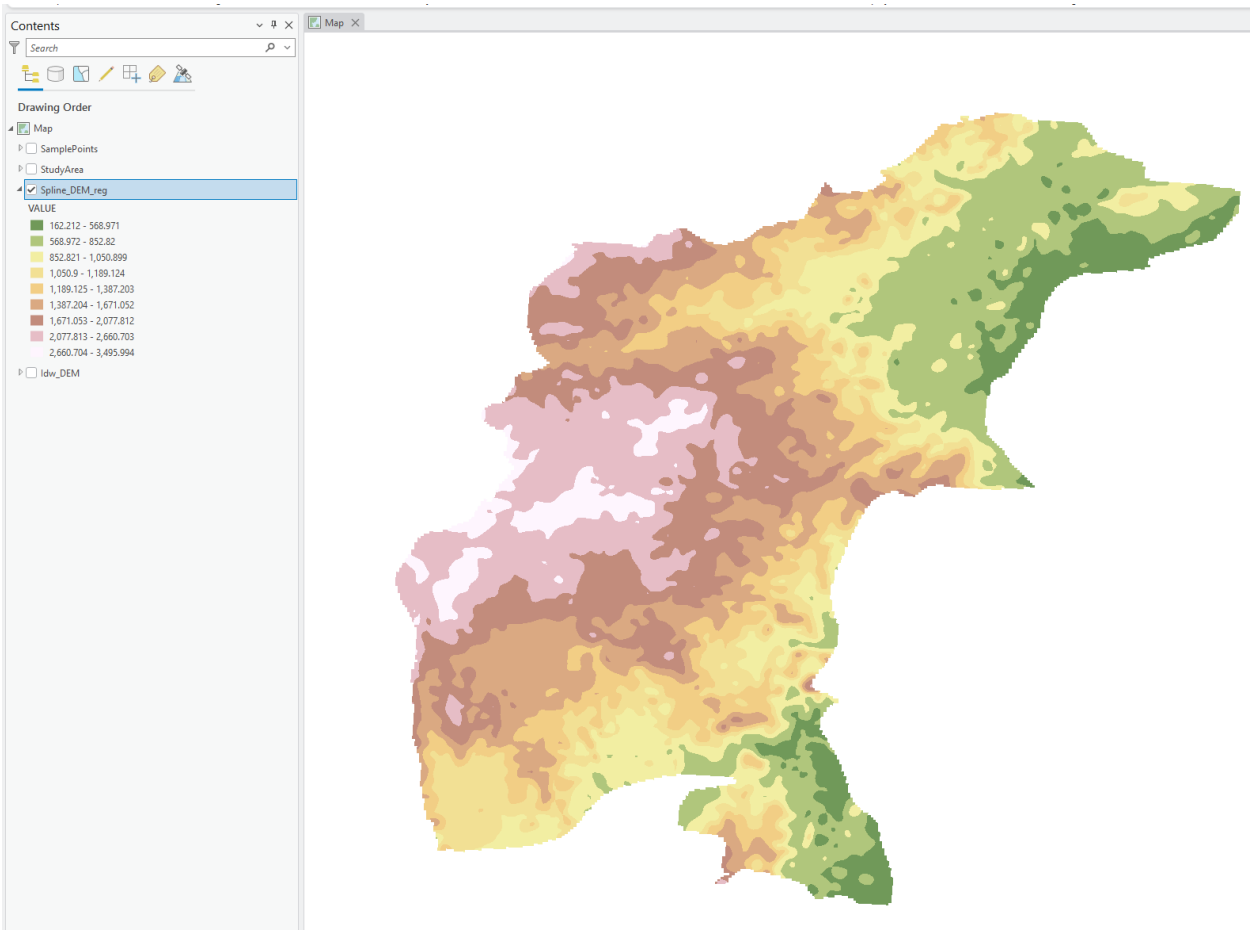
Configure as follows:



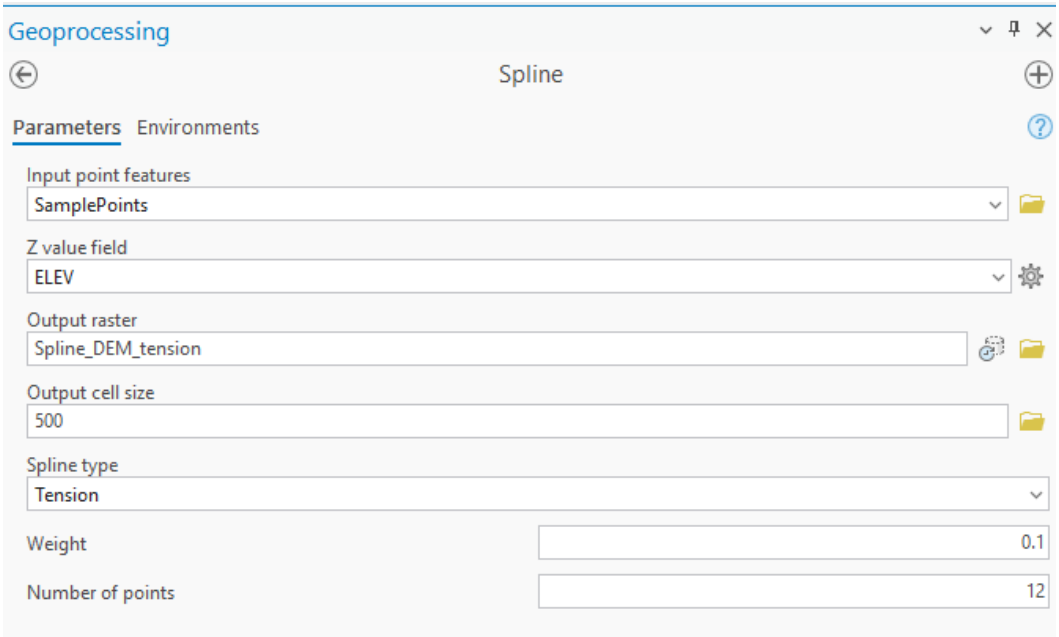
Make sure mask is set:



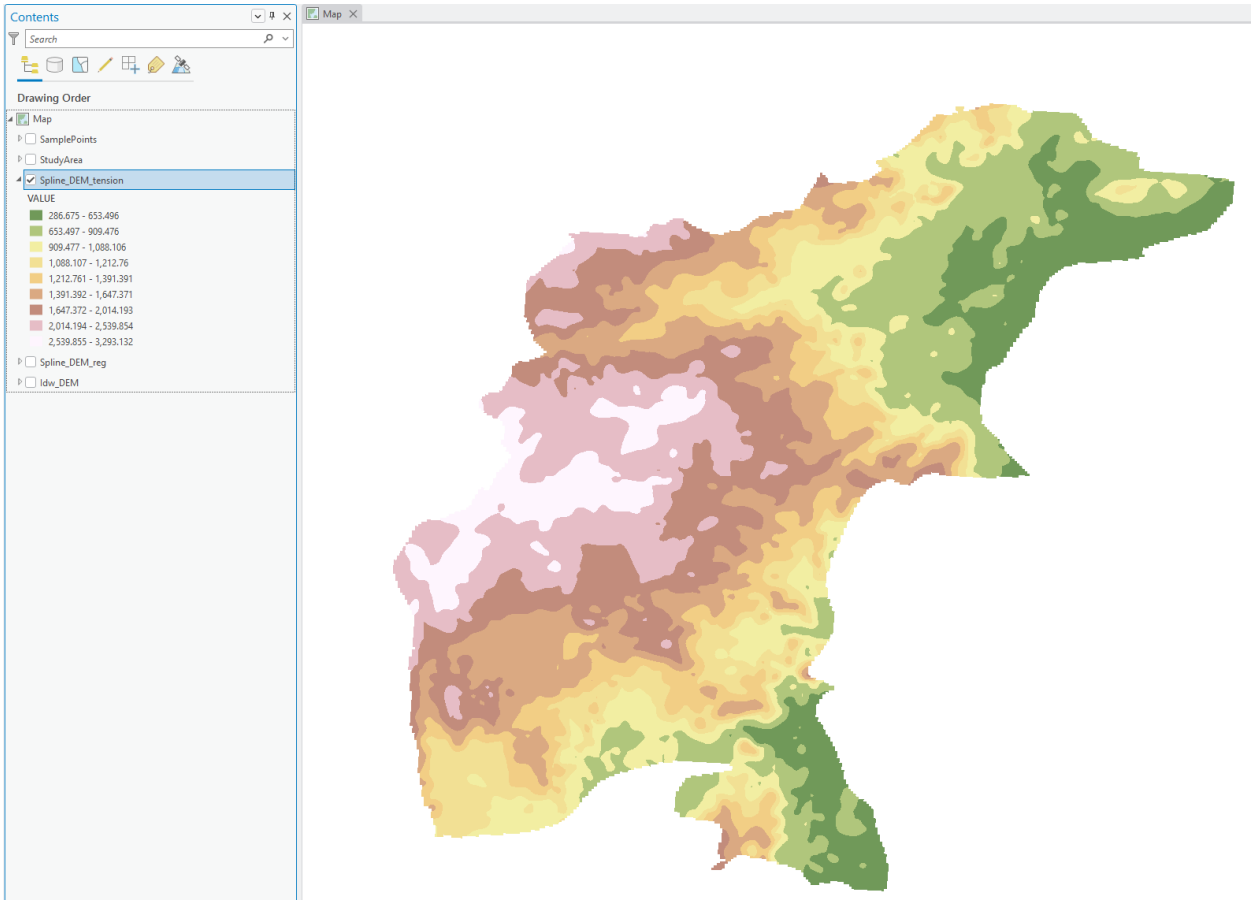
Hit Run:



Run again for tension spline:



Hit **Run:**


Contents
Search
Drawing Order
Map
SamplePoints
StudyArea
Spline_DEM_tension
VALUE
286.675 - 653.496
653.497 - 909.476
909.477 - 1,088.106
1,088.107 - 1,212.76
1,212.761 - 1,391.391
1,391.392 - 1,647.371
1,647.372 - 2,014.193
2,014.194 - 2,539.854
2,539.855 - 3,293.132
Spline_DEM_reg
Idw_DEM
Map

Natural Neighbour Tool

The **Natural Neighbor** tool is an interpolation method that offers a "best of both worlds" approach by being both smooth and highly localized.

It works by creating a **Voronoi (Thiessen) polygon** network around all your input points; to estimate a value for a new location, it temporarily inserts a "hidden" point at that spot and calculates how much area it "steals" from the surrounding polygons.

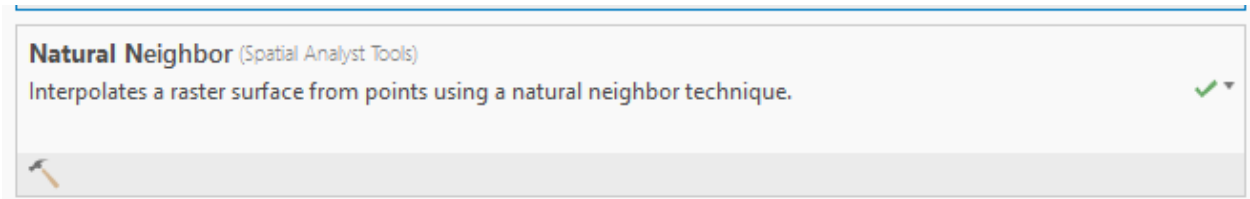
Because it only uses the immediate neighbors to make this calculation, it doesn't create the "bull's-eye" patterns often seen in IDW, nor does it overshoot the data range like Spline can.

It is especially effective for large datasets where you need a surface that looks natural and smooth but stays strictly within the boundaries of your highest and lowest measured values.

Goal: Use Natural Neighbor tool.

Scenario: We know the elevation of a sample of points in our study area. We need to use it to generate an elevation raster for the entire area.

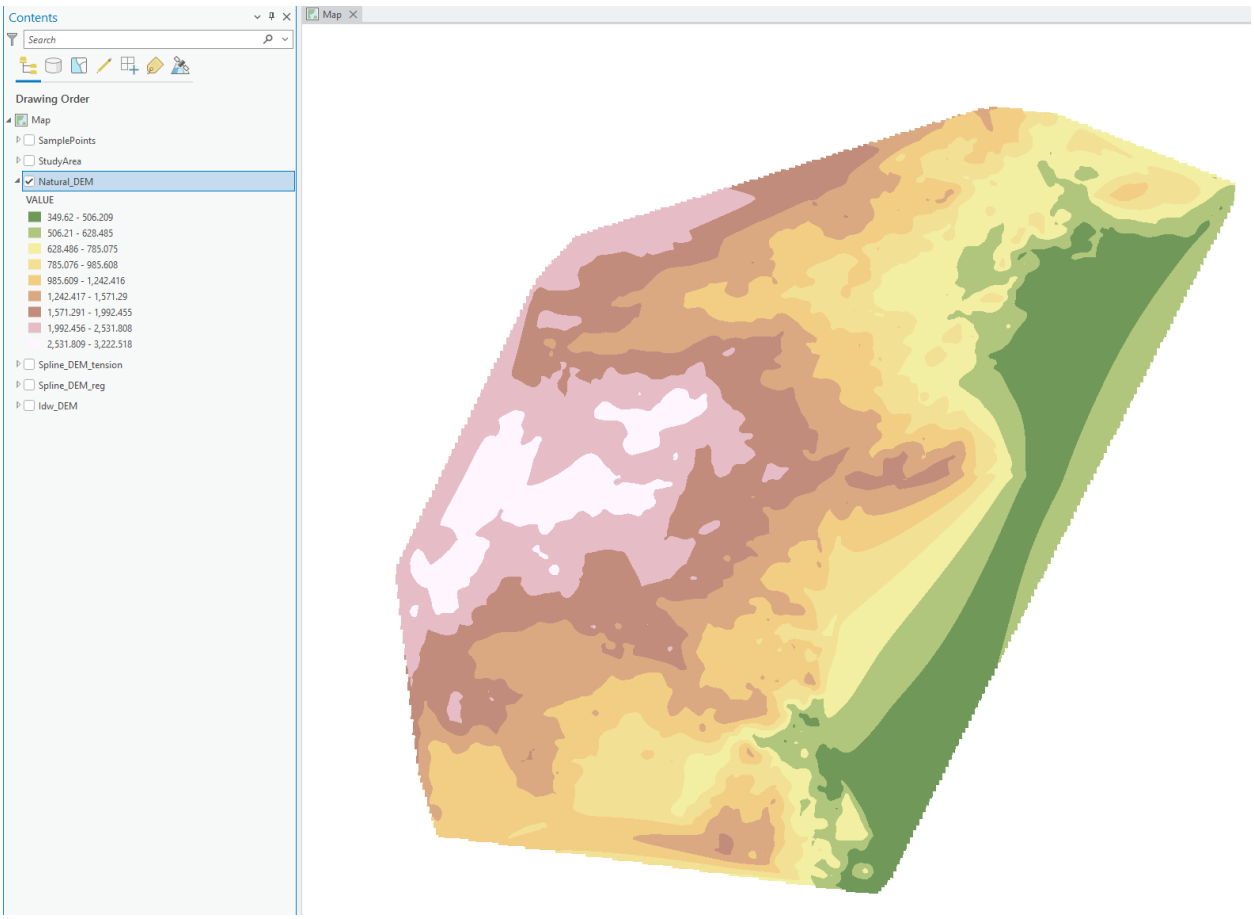
Open Natural Neighbor tool:



Configure as follows:

Geoprocessing
Natural Neighbor
Parameters
Environments
Input point features
SamplePoints
Z value field
ELEV
Output raster
Natural_DEM
Output cell size
500

Hit **Run:**


Contents
Search
Drawing Order
Map
SamplePoints
StudyArea
Natural_DEM
VALUE
349.62 - 506.209
506.21 - 628.485
628.486 - 785.075
785.076 - 985.608
985.609 - 1,242.416
1,242.417 - 1,571.29
1,571.291 - 1,992.455
1,992.456 - 2,531.808
2,531.809 - 3,222.518
Spline_DEM_tension
Spline_DEM_reg
Idw_DEM
Map

Krigging Tool

Kriging is a powerful geostatistical interpolation method that differs from IDW or Spline by utilizing the statistical relationships between measured points to predict unknown values.

Rather than just looking at distance, it uses a **variogram** to account for spatial correlation, the idea that the distance and orientation between points reflect a specific pattern in the landscape.

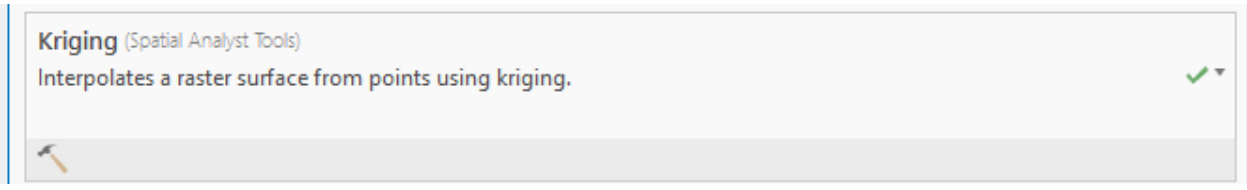
It provides not only an estimated surface but also a measure of certainty or **prediction error**, telling you how much you should trust the results in different areas.

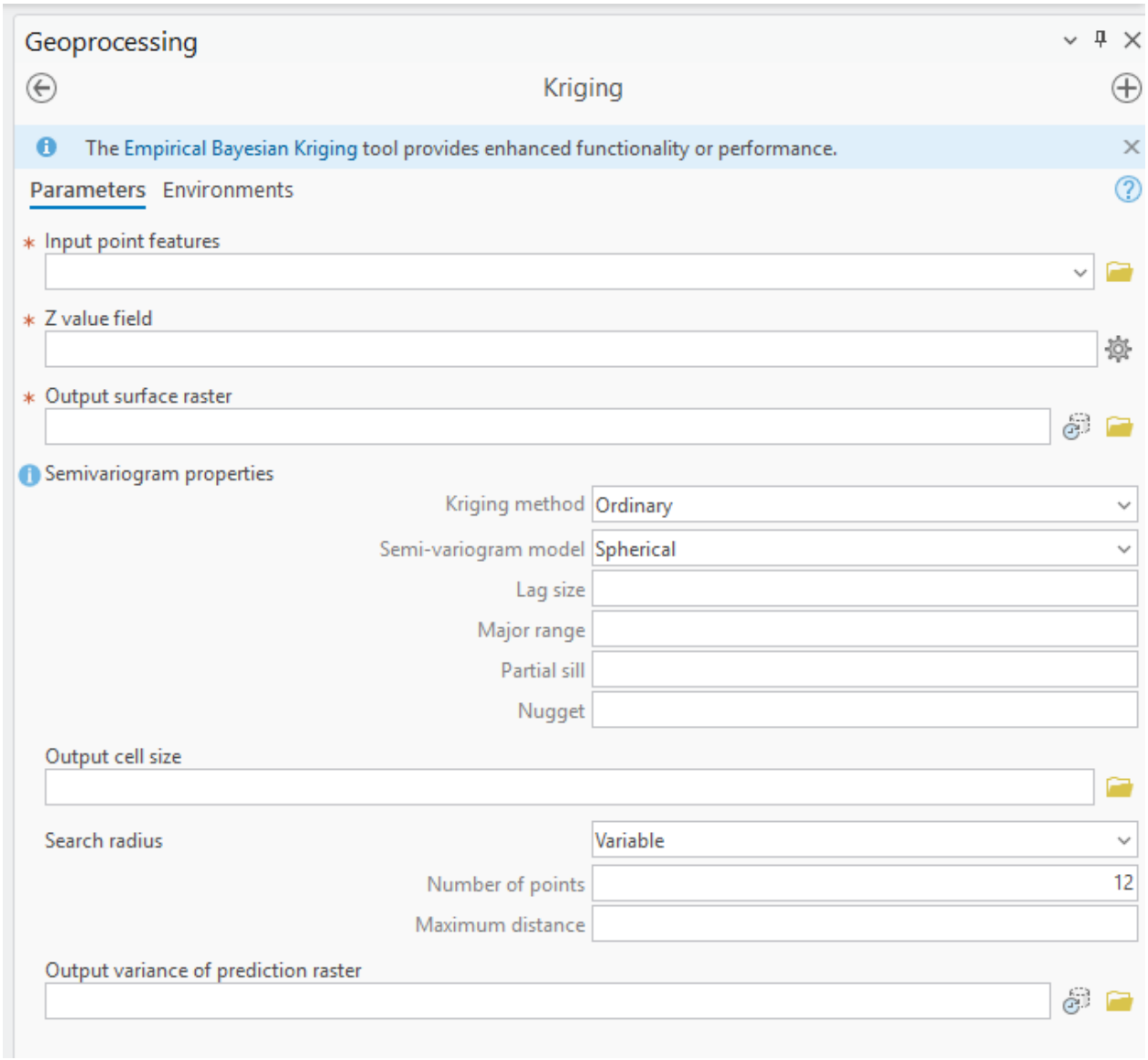
This makes it the preferred tool for complex scientific modeling, such as mining, soil mapping, or environmental science, where understanding the reliability of the map is just as important as the map itself.

Goal: Use Krigging tool.

Scenario: We know the elevation of a sample of points in our study area. We need to use it to generate an elevation raster for the entire area.

Open Krigging Tool:



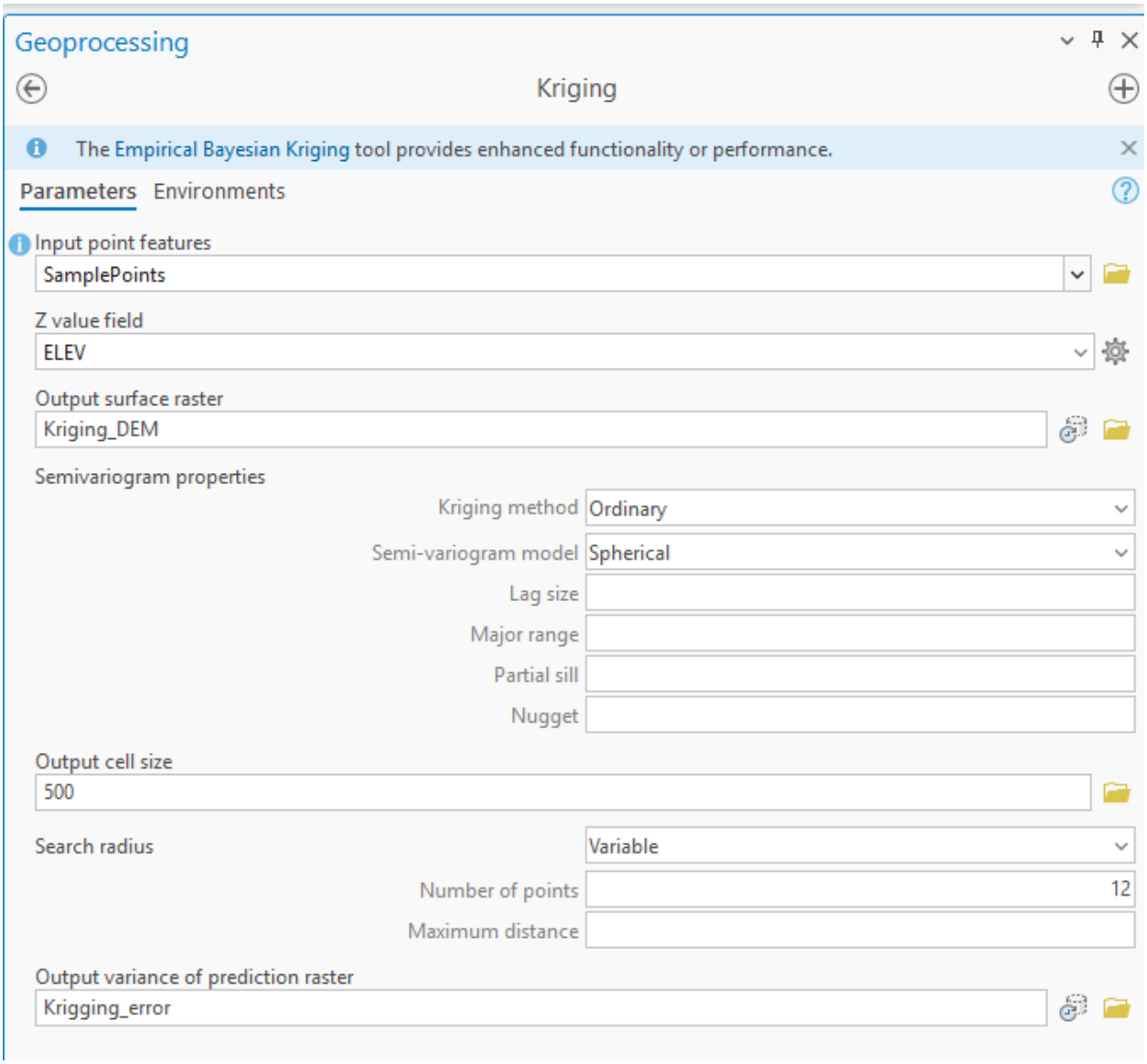


- **Input point features:** The known data points you are using to build the statistical model.
- **Z value field:** The specific measurement (e.g., soil pH or mineral density) you want to map.
- **Output raster:** The final estimated surface file.
- **Output cell size:** The resolution of the final map.

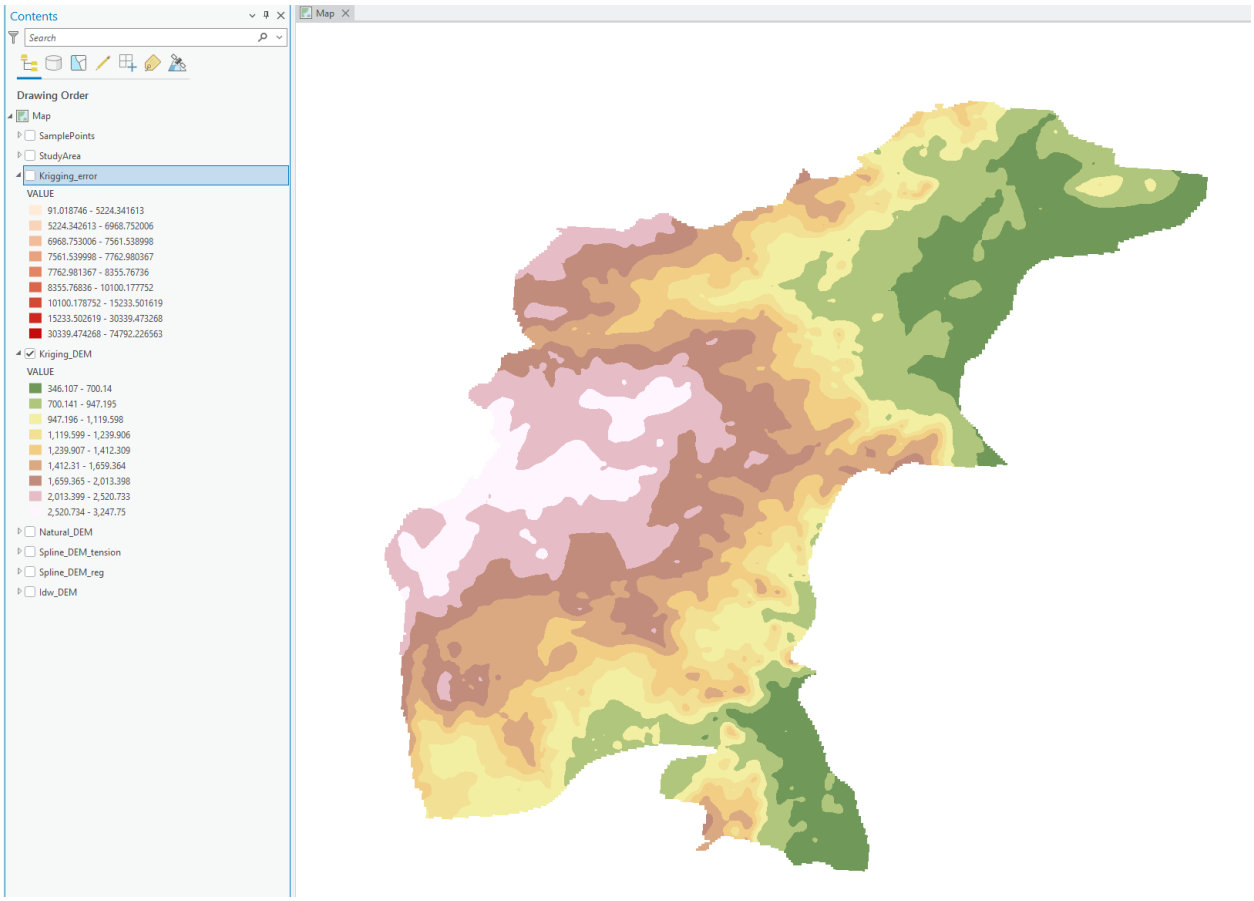
- **Kriging Method:**
    - **Ordinary:** Assumes there is a constant but unknown mean across the data; this is the most common version.

    - **Universal:** Assumes there is an underlying trend (like a steady slope) in the data that needs to be accounted for.
- **Semivariogram Model:** This is the "brain" of Kriging. It defines the mathematical curve (Circular, Spherical, Exponential, etc.) that best describes how the similarity between points drops off as distance increases.
- **Search Radius:**
    - **Variable:** Looks for a set number of points (default is 12) to calculate the value of a cell.
    - **Fixed:** Only looks within a specific distance, regardless of how many points are there.
- **Output prediction error raster:** A unique Kriging feature that creates a second map showing the "uncertainty" or standard error, highlighting where your data is most and least reliable.

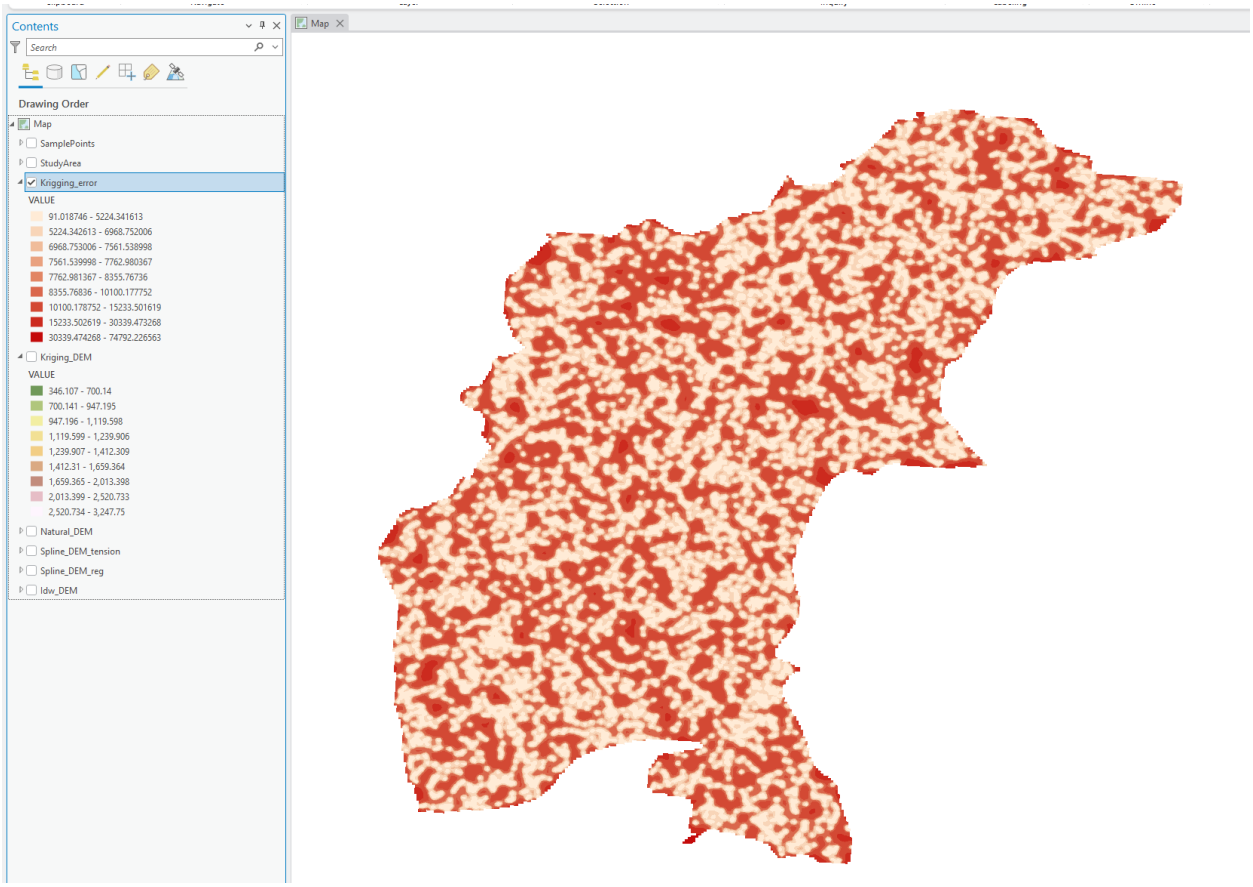
Configure as follows:


Geoprocessing
Kriging
The Empirical Bayesian Kriging tool provides enhanced functionality or performance.
Parameters
Environments
Input point features
SamplePoints
Z value field
ELEV
Output surface raster
Kriging_DEM
Semivariogram properties
Kriging method
Ordinary
Semi-variogram model
Spherical
Lag size
Major range
Partial sill
Nugget
Output cell size
500
Search radius
Variable
Number of points
12
Maximum distance
Output variance of prediction raster
Krigging_error

Hit Run:



The variance raster looks like:



Notice that it’s the least at the points and then increases as we go further.

When to use What?

Choosing the right interpolation method is about matching the mathematical "logic" of the tool to the real-world behavior of what you are mapping.

1. Know Your Boundaries (The "Z-Value" Rule)

- **Use IDW** if your data should never go above or below your measured points. For example, if you are mapping pollution levels, the concentration shouldn't magically become higher than your highest reading between two sensors.
- **Use Spline** if you know the real-world surface has peaks or valleys that your sensors might have missed. Because Spline uses curves, it can "overshoot" the measured points to create a smooth hill or depression.

2. Check Your Data Distribution

- **Use IDW** if you have many points that are close together but have wildly different values. Spline struggles with this and might create strange, unrealistic "waves" in the map.
- **Use Kriging** if your points are sparse or poorly distributed. Kriging is great at handling complex terrain where things change abruptly because it looks at the statistical probability of the data rather than just simple distance.

3. Consider the "Look" and "Certainty"

Method	Best For...	Visual Result
IDW	Simple, quick estimates.	Can look "bumpy" (bull's-eyes).
Spline	Gently varying surfaces (like elevation).	Very smooth and artistic.
Natural Neighbor	Large datasets with varied density.	Natural, localized smoothing.
Kriging	Scientific analysis where accuracy matters.	Statistical and provides "error" maps.

You shouldn't just pick one and hope for the best. The best workflow is to **try multiple methods** and compare them against your own knowledge of the area. If a Spline map shows a massive hill where you know the ground is flat, you'll know to switch to IDW or Kriging.

Interpolation With Barrier Tool

The **Interpolation with Barriers** tool is a specialized version of interpolation method that accounts for physical or logical interruptions in a landscape.

While standard interpolation assumes that values transition smoothly across all space, this tool uses **barrier polyline features** to prevent data from "bleeding" across boundaries where it shouldn't, such as cliffs, rivers, or fault lines.

When the tool encounters a barrier, it essentially "breaks" the connection between points on opposite sides, ensuring that the estimated value for a cell is only influenced by neighbors that have a direct, unobstructed path to it.

This is essential for creating realistic maps in complex environments where a sudden physical change would naturally stop the flow of a variable like water temperature or soil moisture.

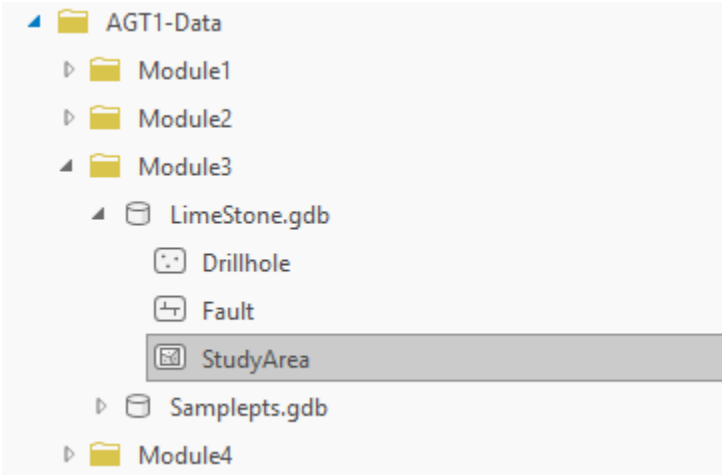
Goal: Use IDW tool with a barrier.

Scenario: Think of an **anticline** like a giant, underground stone bridge or a curved roof. This shape is important because it can act as a natural "container" for oil, gas, or water.

In this task, you are looking at data from drill holes that hit the top of a limestone layer where oil was found. To see how this layer bends underground, you will use the **IDW tool** to turn those specific drill points into a smooth map.

You will run the tool two different ways to see the difference:

1. **Without a Barrier:** A standard map where the data flows freely between all points.
2. **With a Barrier:** A map that accounts for physical breaks (like a fault line) that would stop the rock layer from being continuous.


AGT1-Data
Module1
Module2
Module3
LimeStone.gdb
Drillhole
Fault
StudyArea
Samplepts.gdb
Module4

Contents
Search
Drawing Order
Map
Drillhole
StudyArea
Map

Let’s explore attributes:

Drillhole

Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy

	OBJECTID *	Shape *	API_NUMBER	UTM_EAST	UTM_NORTH	ELEV_M	FORMATION	FORM_NAME	DEPTH_FT	DEPTH_M	SOURCE	SHOW	ELEVTOPM
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2	2	Point	43037100910000	1171353.5669	3645509.8802	1926.0546	359MSSP	Graptolite	9794	2985.2475	DRILLER	FM TEST-O OR G	-1059.19
3	3	Point	43037101560000	1178998.397	3640757.6546	1895.2694	359MSSP	Graptolite	9006	2745.0622	DRILLER		-849.79
4	4	Point	43037103400000	1174424.1692	3639249.4257	1868.4467	359MSSP	Graptolite	8540	2603.0237	DRILLER	FM TEST-O & G	-734.58
5	5	Point	43037113390000	1174041.9226	3644028.0099	2073.5796	359MSSP	Graptolite	9870	3008.4126	DRILLER	FM TEST-O OR G	-934.83
6	6	Point	43037113410000	1172772.5997	3643528.1543	1964.4599	359MSSP	Graptolite	8780	2676.1765	DRILLER	IP TEST-O	-711.72
7	7	Point	43037113420000	1176345.2599	3643079.0387	2002.2555	359MSSP	Graptolite	9660	2944.4038	DRILLER	FM TEST-NO SHOWS	-942.15
8	8	Point	43037113450000	1173538.4355	3648641.8902	2059.2538	359MSSP	Graptolite	9749	2971.5313	DRILLER	CORE-O &/OR G	-912.28
9	9	Point	43037113560000	1182474.4943	3640091.4473	2090.6486	359MSSP	Graptolite	9247	2818.5199	DRILLER	FM TEST-O OR G	-727.87
10	10	Point	43037150490000	1173397.2918	3640732.642	1895.2694	359MSSP	Graptolite	8221	2505.7913	DRILLER	IP TEST-O	-610.52
11	11	Point	43037151230000	1174216.949	3640766.0868	1922.397	359MSSP	Graptolite	8098	2468.3004	DRILLER	IP TEST-O & G	-545.9
12	12	Point	43037157680000	1182963.1649	3639309.3117	2092.4774	359MSSP	Graptolite	9112	2777.3714	DRILLER	IP TEST-G	-684.89
13	13	Point	43037157690000	1177623.6008	3641378.6227	1940.6852	359MSSP	Graptolite	8434	2570.7145	DRILLER	IP TEST-O & G	-630.03
14	14	Point	43037162190000	1173973.5773	3647107.2639	2171.4216	359MSSP	Graptolite	9801	2987.3811	DRILLER	IP TEST-O & G	-815.96
15	15	Point	43037162360000	1177092.0087	3640448.2938	2112.8993	359MSSP	Graptolite	9180	2798.098	DRILLER	FM TEST-O OR G	-685.2
16	16	Point	43037162360001	1177092.0087	3640448.2938	2113.8137	359MSSP	Graptolite	8998	2742.6238	DRILLER	IP TEST-O & G	-628.81
17	17	Point	43037162370000	1173700.7053	3640161.0917	1860.8266	359MSSP	Graptolite	8126	2476.8349	DRILLER	IP TEST-O & G	-616.01
18	18	Point	43037162380000	1175604.696	3640089.7043	1958.059	359MSSP	Graptolite	8806	2684.1014	DRILLER	IP TEST-O	-726.04
19	19	Point	43037162390000	1172617.0137	3641093.0015	1880.6389	359MSSP	Graptolite	8390	2557.3031	DRILLER	IP TEST-O & G	-676.66
20	20	Point	43037162400000	1177664.9872	3640912.1297	1946.4765	359MSSP	Graptolite	8630	2630.456	DRILLER	IP TEST-O & G	-683.98

Don’t worry about the other fields, we only need to know that **ELEVTOPM** shows the depth till the limestone formation.

Open IDW Tool:

IDW (Spatial Analyst Tools)

Interpolates a raster surface from points using an inverse distance weighted (IDW) technique.

Configure as follows (without barrier):

Geoprocessing

IDW

Parameters Environments

Input point features: Drillhole

Z value field: ELEVTOPM

Output raster: Idw_Drillhol1

Output cell size: 50

Power: 2

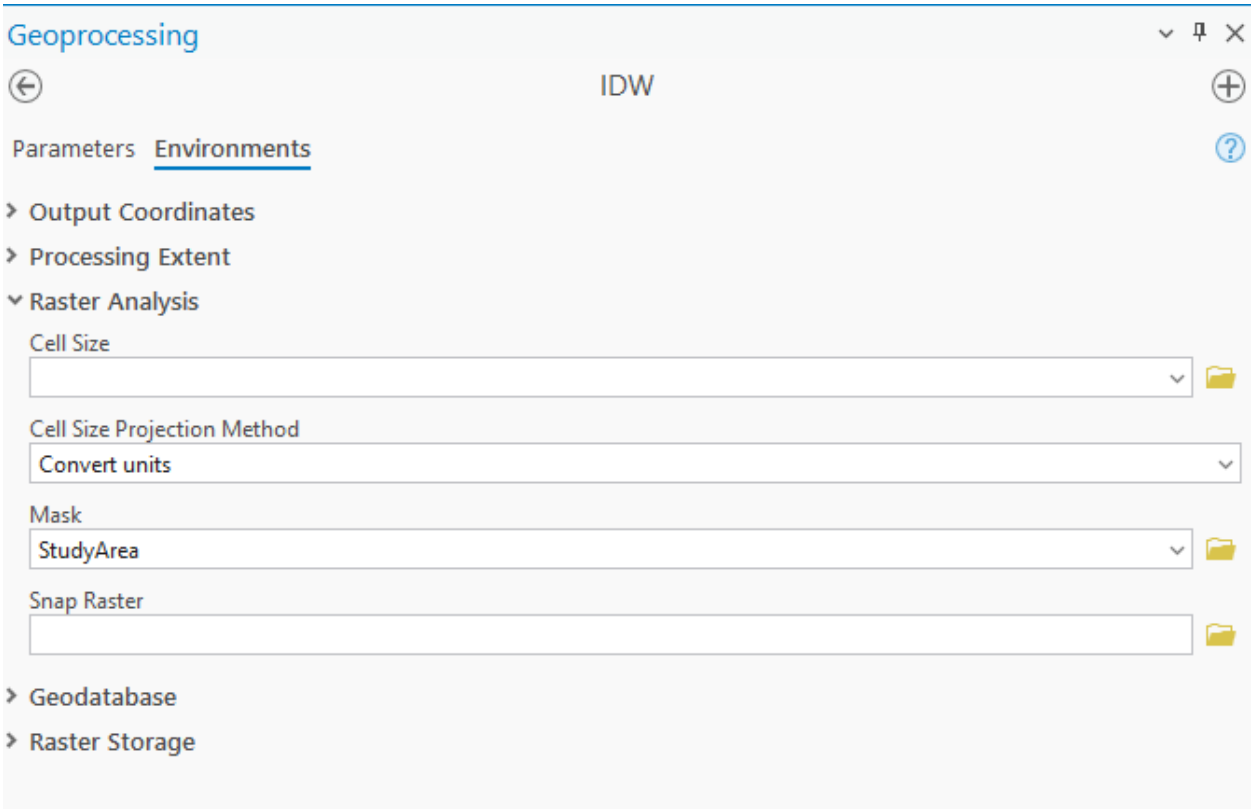
Search radius: Variable

Number of points: 12

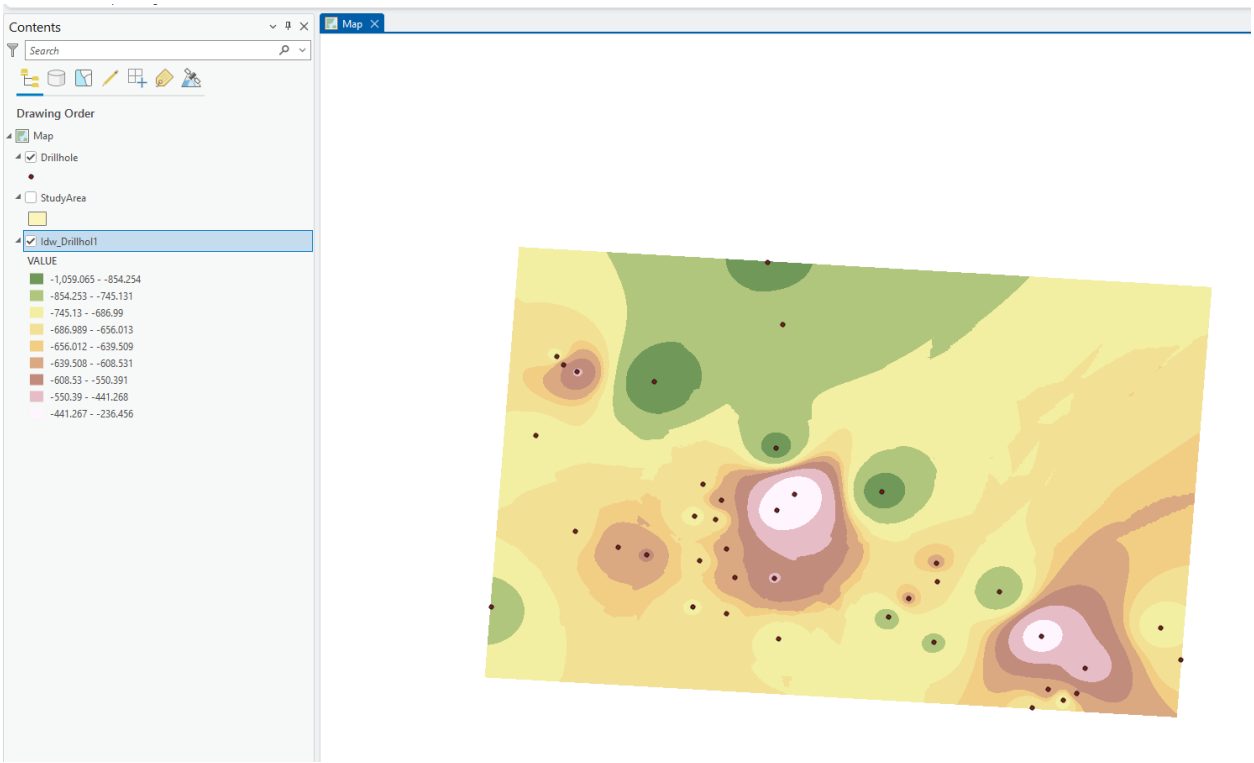
Maximum distance:

Input barrier polyline features:

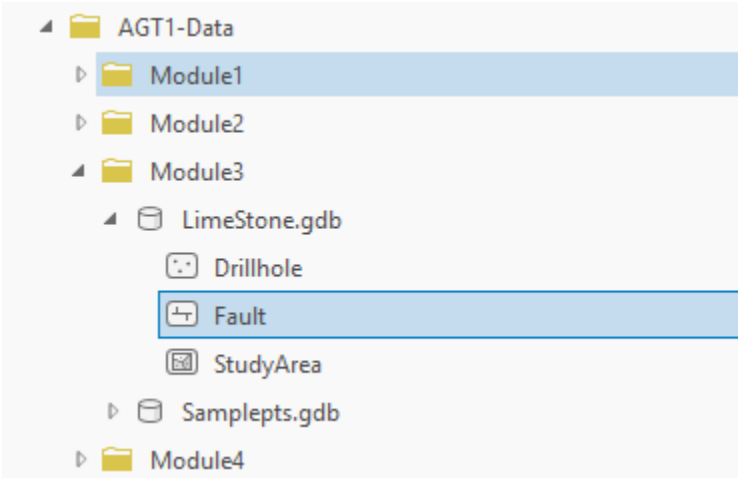
Ensure mask is set to study area:

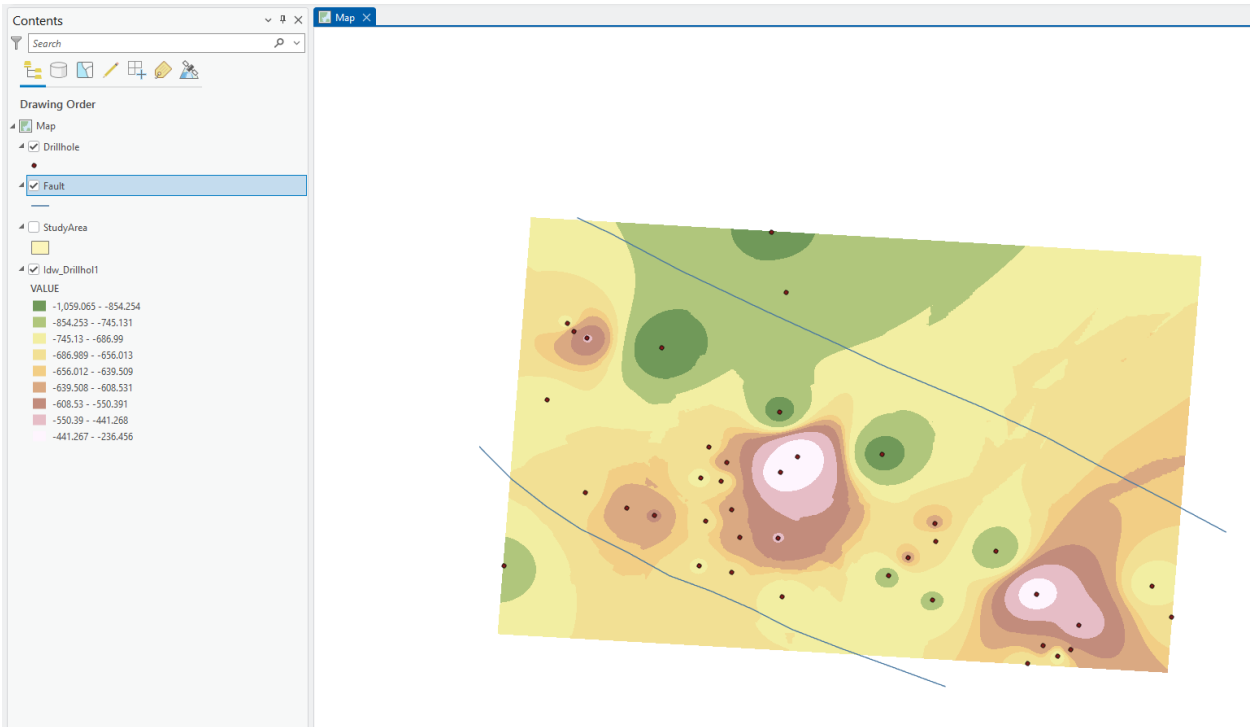


Hit **run**:

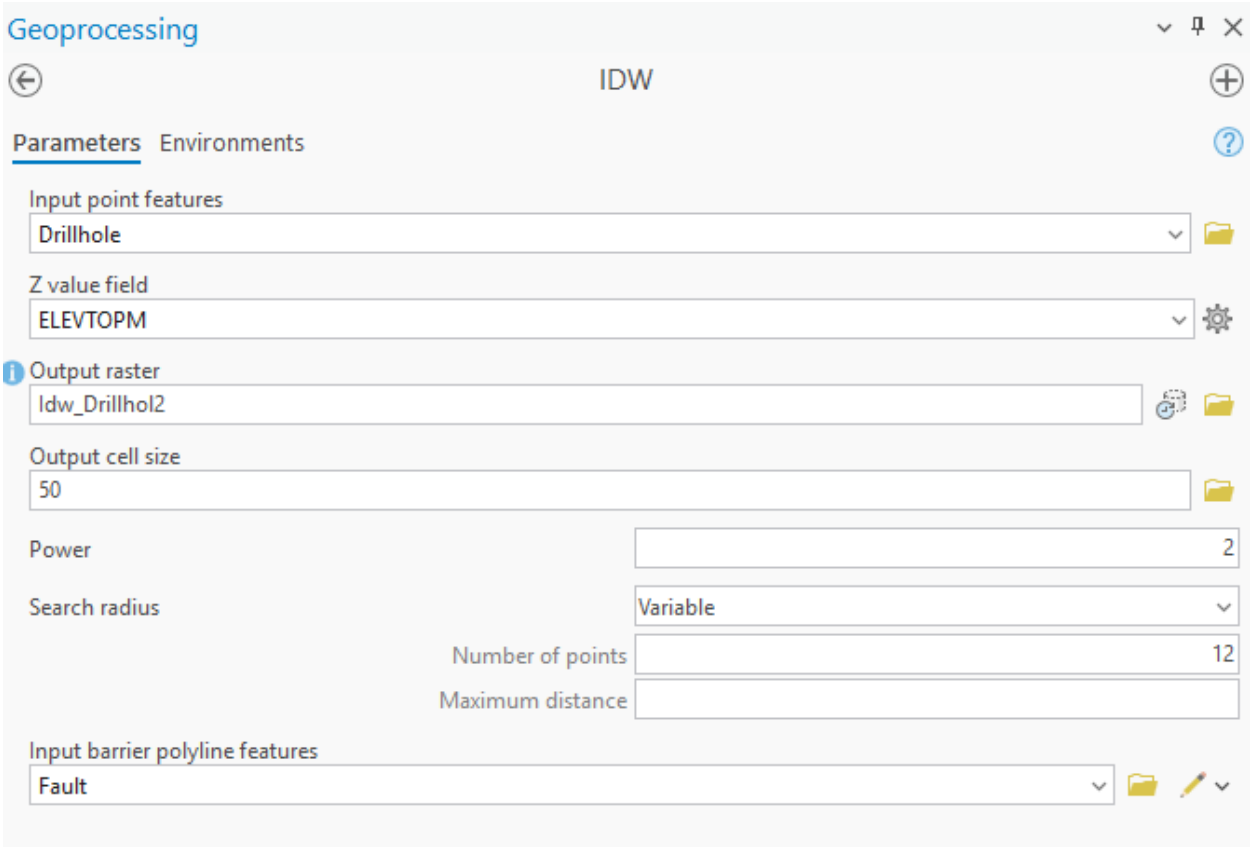


Now load **Fault Lines:**

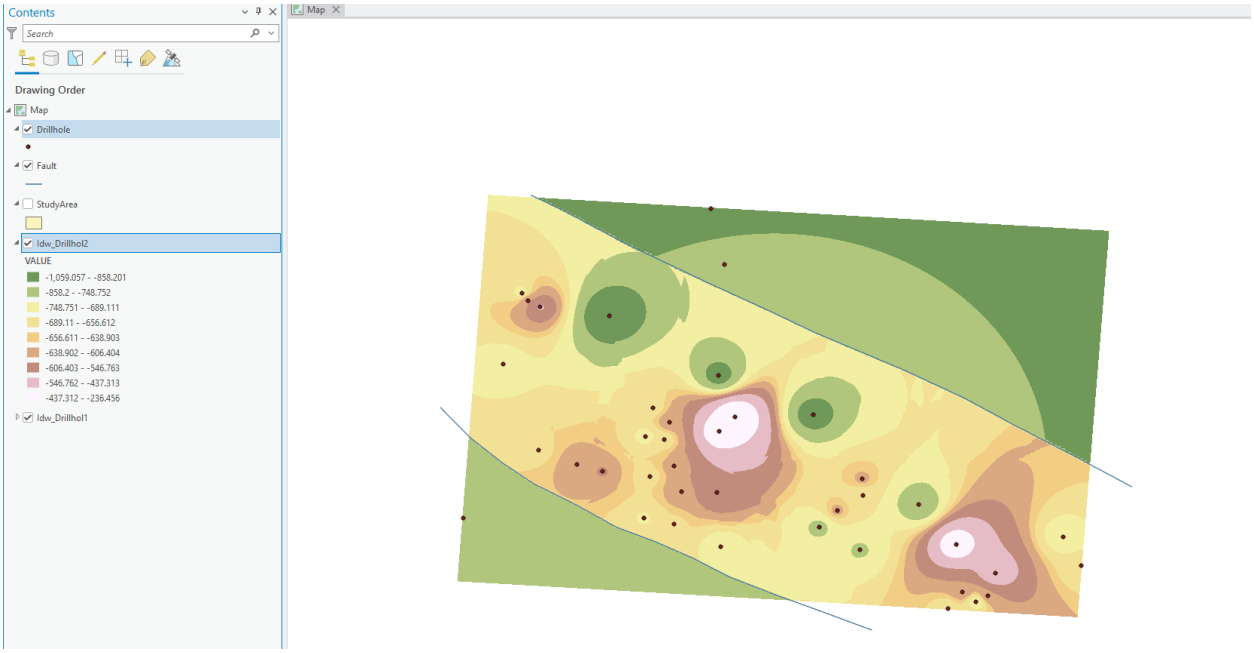




Re-run the IDW tool but this time set **Fault** as the interpolation barrier:

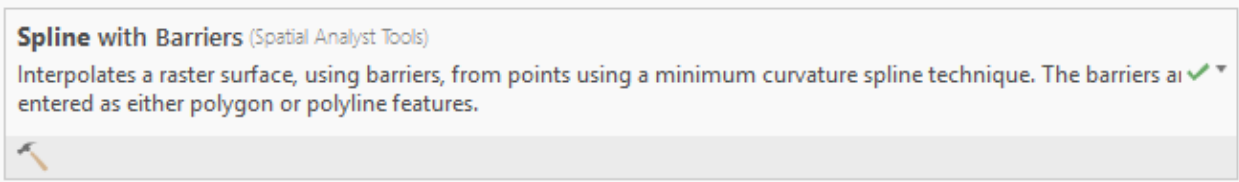


Hit Run:

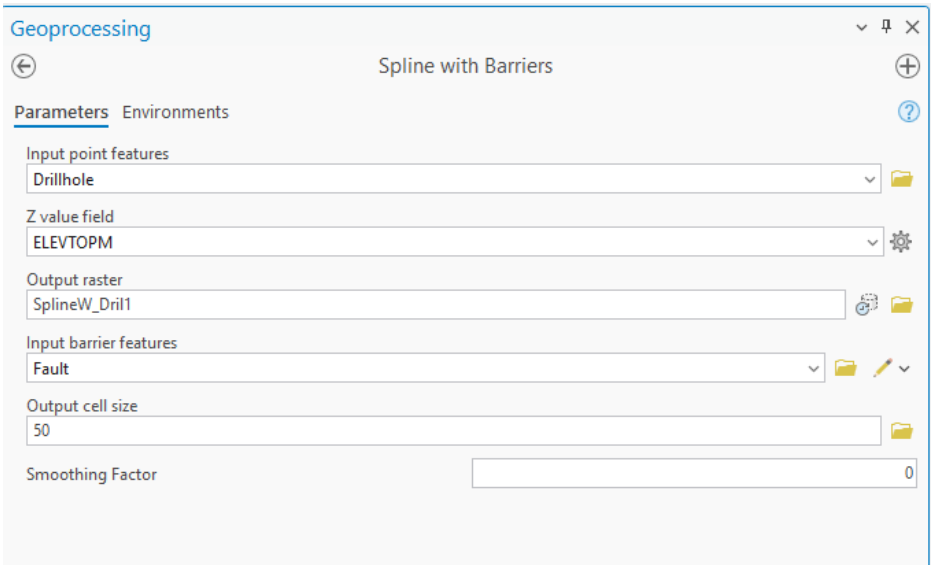


You can notice a clear difference due to the fault line.

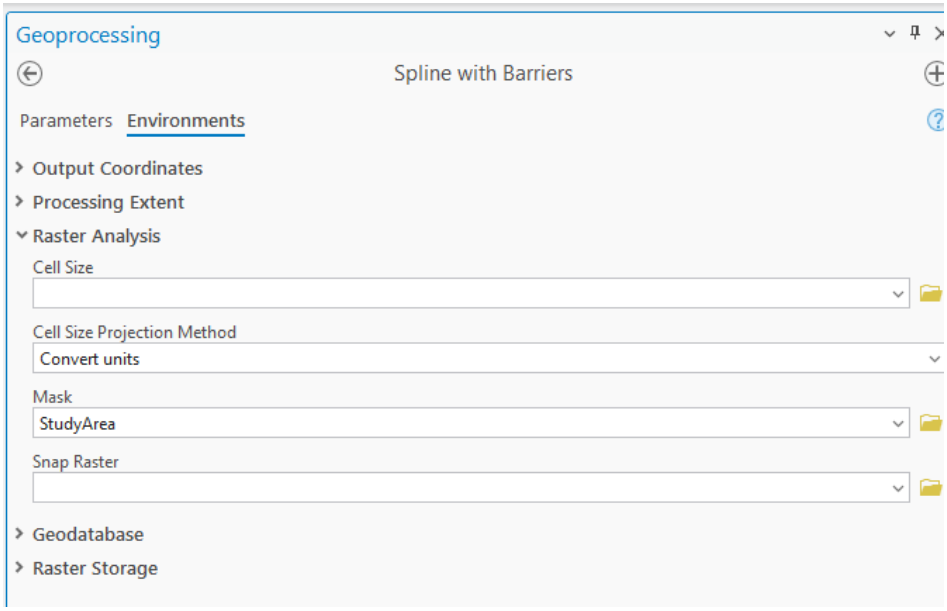
We can also use spline with barrier:



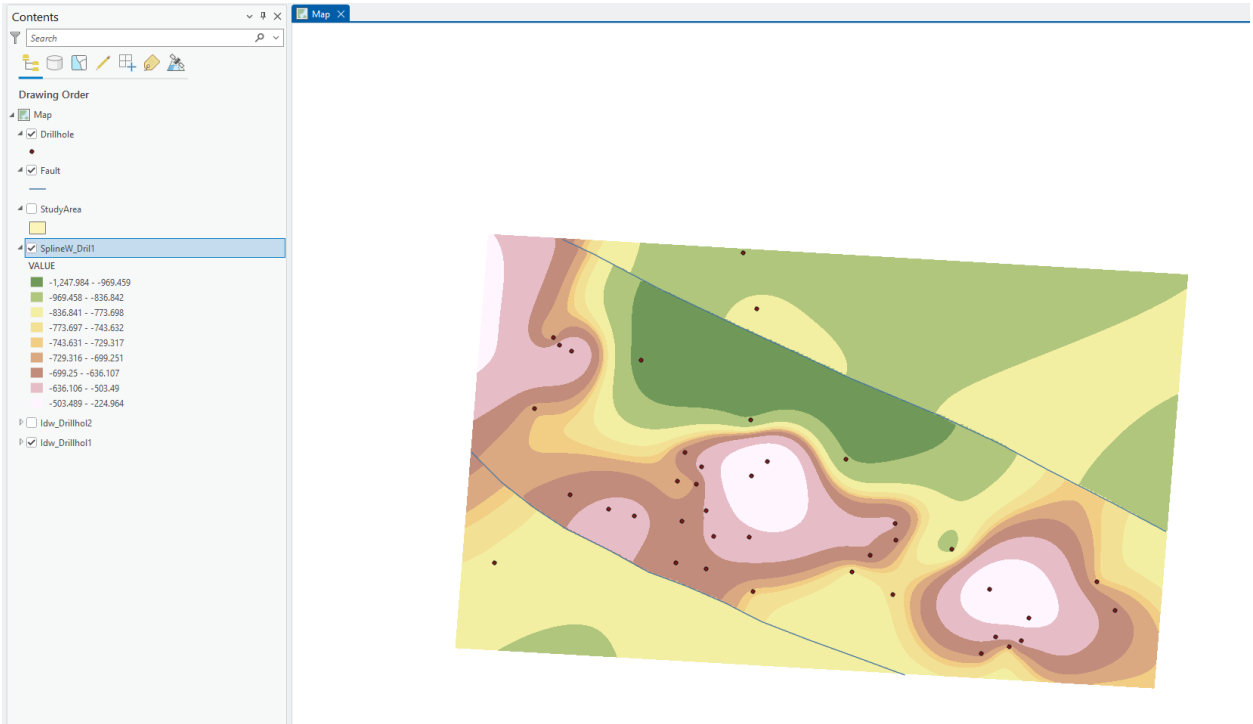
Configure as follows:



Ensure Mask is set:



Hit **run:**


Contents
Search
Drawing Order
Map
Drillhole
Fault
StudyArea
SplineW_Dril1
VALUE
-1,247.984 - -969.459
-969.458 - -836.842
-836.841 - -773.698
-773.697 - -743.632
-743.631 - -729.317
-729.316 - -699.251
-699.25 - -636.107
-636.106 - -503.49
-503.489 - -224.964
Idw_Drillhol2
Idw_Drillhol1
Map

Deliverables (To be submitted)

- Map showing use of each IDW tool. (Screenshot **not** Map Layout)
- Map showing use of both spline tools. (Screenshot **not** Map Layout)
- Map showing use of Kriging tool. (Screenshot **not** Map Layout)
- Map showing use of IDW tool with barrier. (Screenshot **not** Map Layout)
- Map showing use of spline with barrier tool. (Screenshot **not** Map Layout)